Lecture 29: Divergence theorem (cont.): applications and proof

$$\oiint_S \vec{F}\cdot d\vec{S} = \iiint_D \text{div}\,\vec{F}\,dV \iff \oiint_S \langle P,Q,R\rangle\cdot\hat{n}\,dS = \iiint_D (P_x + Q_y + R_z)\,dV$$

∇ notation: ∇ "delta" $= \langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\rangle$

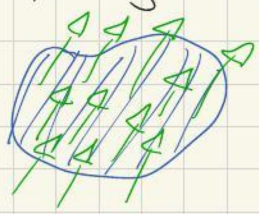
$\nabla f = \langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\rangle$ gradient

$\nabla\cdot\vec{F} = \langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\rangle \cdot \langle P,Q,R\rangle = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$ = Divergence

Physical Interpretation: $\text{div}(\vec{F})$ = "source rate" = amount of flux generated per unit volume

If we have an incompressible fluid flow: (given mass always occupies same volume) $\Rightarrow$ Velocity $\vec{F}$

Then The Divergence thm says: "If i sum the Divergence in D, im going to get the flux going out through this surface"



Reminder: Flux: How much fluid is going through this surface

$\iiint_D \text{div}\,\vec{F}\,dV = \oiint \vec{F}\cdot\hat{n}\,d$ = amount of fluid leaving D per unit time

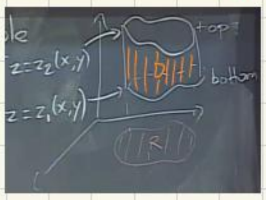
"Integrating the divergence gives you the total amount of sources − the total amount of sinks"

Divergence mesures the amount of sources/sinks per unit volume

$$\boxed{\text{Proof of } \oiint_S \langle 0,0,R\rangle\cdot\hat{n}\,dS = \iiint_D R_z\,dV}$$

(Then get the general case by summing one such identity for each component)

* If region D is vertically simple



Right-hand side: $\iiint_D R_z\,dV = \iint_R \left(\int_{z_1(x,y)}^{z_2(x,y)} R_z\,dz\right)dx\,dy = \iiint_D R_z\,dV = \iint_R \left[R(x,y,z_2(x,y)) - R(x,y,z_1(x,y))\right]dx\,dy$

The other side: $\oiint_{\substack{S\\ \text{bottom+top+sides}}} \langle 0,0,R\rangle\cdot\hat{n}\,dS = \iint_{Top} + \iint_{bottom} + \iint_{sides}$

Top: graph $z = z_2(x,y)$ then $\hat{n}\,dS = \langle -\frac{\partial z_2}{\partial x}, -\frac{\partial z_2}{\partial y}, 1\rangle\,dx\,dy$

$\langle 0,0,R\rangle\cdot\hat{n}\,dS = R\,dx\,dy$ so $\iint_{top}\langle 0,0,R\rangle\cdot\hat{n}\,dS = \iint_{top} R\,dx\,dy = \iint_R R(x,y,z_2(x,y))\,dx\,dy$

Bottom: graph $z = z_1(x,y)$

$\underbrace{\hat{n}\,dS = \langle \frac{+\partial z_1}{\partial x}, \frac{+\partial z_1}{\partial y}, -1\rangle\,dx\,dy}_{\text{with } \hat{n} \text{ pointing downwards}}$ $\iint_{bottom}\langle 0,0,R\rangle\cdot\hat{n}\,dS = \iint_{bottom} -R\,dx\,dy = \iint_R -R(x,y,z_1(x,y))\,dx\,dy$

For the sides, since the vector field is parallel to the z-axis, nothing much happens on the sides, no fluid is going out/in through the sides.

Sides are vertical, $\langle 0,0,R\rangle$ is tangent to sides, flux through sides = 0

So: $\iiint_D R_z\,dV = \oiint_{\text{Top+bottom+sides}} \langle 0,0,R\rangle\cdot\hat{n}\,dS$

* If D isn't vertically simple: We cut it in vertically simple pieces

Lecture 29 (part 2)

Diffusion equation: = Heat Equation:	When you have a fluid and introduce some substance and want to figure out how it's going to spread out/diffuse into the ambient fluid Governs the motion of smoke in immobile air or dye in a solution

unknown

u = concentration at a given point, therefore depending on where you are, $u(x,y,z,t)$. It will give us $\frac{\partial u}{\partial t}$, "how the concentration at a given point varies over time in terms of how the concentration varied in space"

= (temperature)

$$\frac{\partial u}{\partial t} = k \underbrace{\nabla^2 u}_{\text{Laplacian of } u} = k \underbrace{\nabla \cdot \nabla u}_{\text{div}(\text{grad}(u))} = k\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right)$$

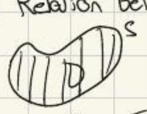
$\vec{F}$ = flow of smoke

1) Physics (+ common sense): smoke flows from high concentration areas to low concentration areas

"Always going in the direction where the concentration decreases the fastest" Therefore negative gradient

So $\vec{F}$ is directed along $-\nabla u$ In fact: $\underbrace{\vec{F} = -k\nabla u}_{\text{proportional}}$ The bigger the difference or "gap" the more intense the flow will be

2) Relation between $\vec{F}$ and $\frac{\partial u}{\partial t}$? Divergence theorem



$$\iiint_D \text{div}\vec{F}\, dV = \oiint_S \vec{F}\cdot\hat{n}\, dS = \text{amount of smoke through } S \text{ per unit time} = -\frac{\partial}{\partial t}\left(\iiint_D u\, dV\right)$$

$$\text{So } \iiint_D \text{div}\vec{F}\, dV = -\frac{\partial}{\partial t}\left(\iiint_D u\, dV\right) = -\iiint_D \frac{\partial u}{\partial t}\, dV$$

$$\frac{\partial}{\partial t}\left(\sum_i u(x_i, y_i, z_i, t)\, \Delta V_i\right) = \sum \frac{\partial u}{\partial t}(x_i, y_i, z_i, t)\, \Delta V_i$$

For any region D

$$\frac{1}{\text{vol}(D)} \iiint_D \text{div}\vec{F}\, dV = \frac{1}{\text{Vol} D} \iiint_D -\frac{\partial u}{\partial t}\, dV \iff \text{The average of } (\text{div}\vec{F}) \text{ in } D = \text{average of } \left(\frac{\partial u}{\partial t}\right) \text{ in } D \Rightarrow \text{div}\vec{F} = -\frac{\partial u}{\partial t}$$

$$\frac{\partial u}{\partial t} = -\text{div}\vec{F} = +k\,\text{div}(\nabla u) = k\nabla^2 u \iff \frac{\partial u}{\partial t} = k\nabla^2 u \text{ diffusion equation}$$